Hold On to Your Cash:

CEO Speech Uncertainty and Financing Conservatism

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Abstract

[Abstract pending Step 6 prose population.]

I Introduction

Earnings calls are a high-frequency setting where CEOs describe firm conditions to outsiders in their own words. We ask whether CEO speech uncertainty during these calls predicts firm precautionary cash holdings. Our measurement is built on the CEO speech-uncertainty method of Dzieliński, Wagner, and Zeckhauser (2021), which we extend from market reactions, analyst behavior, firm performance, and governance outcomes to firm financing-policy outcomes.

We document that within-firm increases in CEO speech uncertainty during earnings calls predict higher precautionary cash holdings, identifying a real-time signal of firm financing-policy adjustment grounded in the precautionary cash motive (Opler, Pinkowitz, Stulz, & Williamson, 1999; Bates, Kahle, & Stulz, 2009). The relation is positive in twelve of twelve specifications (six contemporaneous and six one-quarter-ahead) under one-tailed tests at the ten-percent level. The cash response amplifies robustly under financing-friction stress and conditionally under cash-flow-volatility stress, consistent with a single precautionary channel triggered by two distinct stress conditions (Almeida, Campello, & Weisbach, 2004; Han & Qiu, 2007).

External uncertainty proxies—political risk, economic policy uncertainty, and product-market competition—covary with the speech-uncertainty measure in the predicted directions, supporting the measure's construct validity. Post-call bid-ask spreads widen and SEC conference-call comment letters reference the same speech signal, corroborating the precautionary interpretation through market information-asymmetry and disclosure-insufficiency channels.

Reverse-causality concerns are addressed via cross-sectional modus tollens on the two precautionary triggers, and we acknowledge as a limitation that no exogenous-shock or instrumental-variable identification is pursued.

Our main contribution is to establish CEO speech uncertainty during earnings calls as a real-time signal of firm precautionary cash holdings, extending the method beyond its original scope of market reactions, analyst behavior, firm performance, and governance outcomes. The remainder of the paper proceeds as follows. Section II develops the conceptual framework, hypotheses, and empirical strategy. Section III presents the main empirical analyses—cash holdings, the two precautionary triggers, and the endogeneity treatment. Section IV extends to outsider-reaction tests. Section V concludes with limitations and directions for future work.

II Conceptual Framework and Empirical Strategy

II.1 Conceptual Framework

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II.2 Hypothesis Development

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II.3 Estimation of Main Variables

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II.4 Methodology

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II.5 Specification and Measurement of Key Constructs

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III Main Empirical Analyses

III.A Data, Sample, and Variable Construction

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III.B Cash Holdings (HC)

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III.C Constraint Amplification (HFC)

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III.D Cash-Flow-Volatility Amplification

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III.E Endogeneity

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IV Additional Analyses

IV.A Market Information-Asymmetry Channel: Post-Call Bid-Ask Spread

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IV.B Disclosure-Insufficiency Channel: Conference-Call Comment Letter (CCCL)

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V Conclusion

V.1 Summary

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V.2 Limitations

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V.3 Future Work

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A Variable Definitions

This appendix lists every variable used in the empirical analyses, with construction formula, data source, and originating reference. Variable names follow paper-standard conventions where available (Dzieliński et al. 2021 for speech measures, Bates et al. 2009 for cash-holdings controls); deviations from source-paper frequency or sample are documented in the formula column.

A.1 Sample Manifest

Name	Description / Formula	Source	Reference
`file_name, ceo_id, ceo_name, gvkey, ff12_code, ff12_name, start_date`	Master sample manifest with CEO, firm, and industry identifiers	outputs/1.4_AssembleManifest	

A.2 Text/Linguistic Variables

Name	Description / Formula	Source	Reference
`ClarityCEO`	CEO clarity persistent-style component (DWZ Eqn 5). *Formula:* ClarityCEO$_i$ = $-\widehat{\mathrm{FE}}_i$ where $\widehat{\mathrm{FE}}_i$ is the CEO-i fixed-effect estimate from the panel regression UncAnsCEO$_{ic} = \alpha + \sum_i \mathrm{FE}_i \cdot \mathbb{1}[\mathrm{CEO} = i] + \mathbf{X}_{ic}\gamma + \varepsilon_{ic}$ over the call-level Q&A uncertainty panel; controls $\mathbf{X}$ are the DWZ extended-controls vector. The sign flip on $\widehat{\mathrm{FE}}_i$ makes higher ClarityCEO mean "CEO speaks more clearly" (lower mean uncertainty). Persistent CEO trait.	outputs/econometric/ceo_clarity_extended/	Dzielínski et al. (2021) (Section 4.4 Eqn 5, p.16)
`ClarityCEO_QtrExp`	ClarityCEO under within-tenure expanding window. *Formula:* As ClarityCEO, but the CEO fixed-effect regression is re-estimated on an expanding within-CEO window using only calls of CEO i available up to and including focal call c. Strictly look-ahead-free.	outputs/econometric/ceo_clarity_expanding/	thesis QtrExp variant of Dzielínski et al. (2021) Eqn 5
`NegCall`	Entire-call negative sentiment percentage. *Formula:* Ratio of negative words to total words in the entire call, based on LM 2011 negative-words list. Per DWZ 2021 Appendix Table A.1: 'percentage of negative words in all words spoken by the CEO, CFO and analysts attending the call.'	outputs/2_Textual_Analysis/2.2_Variables	Dzielínski et al. (2021) (Section 4.4, p.18; Appendix Table A.1, p.54)
`UncAnsCEO`	CEO-only Q&A uncertainty percentage (input to ClarityCEO/UncResCEO residualization). *Formula:* CEO Q&A uncertainty (%) = UnctWordsAnsCEO / WordsAnsCEO (DWZ 2021 Eqn 2). Uncertainty wordlist = LM 2011 Master Dictionary 'uncertainty' list (297 words, Aug 2014 version). CEO identified via Execucomp ceoann field. Used in body only as the dependent variable inside the CEO fixed-effect residualization that produces ClarityCEO and UncResCEO.	outputs/2_Textual_Analysis/2.2_Variables	Dzielínski et al. (2021) (Section 4.2, Eqn 2, p.13)
`UncPreCEO`	CEO-only presentation-segment uncertainty percentage (DWZ Eqn 1). *Formula:* CEO Presentation uncertainty (%) = UnctWordsPreCEO / WordsPreCEO. Uncertainty wordlist = LM 2011 (297 words).	outputs/2_Textual_Analysis/2.2_Variables	Dzielínski et al. (2021) (Section 4.2, Eqn 1, p.13)
`UncQue`	Analyst Q&A uncertainty percentage (DWZ Eqn 3). *Formula:* Analyst Q&A uncertainty (%) = UnctWordsQue / WordsQue. Uncertainty wordlist = LM 2011 (297 words). Used as ΔUncQue control in DWZ first-difference replication.	outputs/2_Textual_Analysis/2.2_Variables	Dzielínski et al. (2021) (Section 4.2, Eqn 3, p.13)

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Name	Description / Formula	Source	Reference
`UncResCEO`	CEO Q&A residual call-state uncertainty component (DWZ Eqn 4). *Formula:* $\widehat{\varepsilon}_{ic}$ = the residual from the panel regression UncAnsCEO$_{ic}$ = $\alpha + \sum_i \mathrm{FE}_i \cdot \mathbb{1}[\mathrm{CEO} = i] + \mathbf{X}_{ic}\boldsymbol{\gamma} + \varepsilon_{ic}$ that produces ClarityCEO. By OLS first-order conditions, the within-CEO mean of UncResCEO is exactly zero by construction. Captures call-level deviation in CEO Q&A uncertainty after stripping out the persistent CEO mean.	outputs/econometric/ceo_clarity_extended/	Dzieliński et al. (2021) (Section 4.4 Eqn 4, p.16)
`UncResCEO_QtrExp`	UncResCEO under within-tenure expanding window. *Formula:* As UncResCEO, but the residualization regression is re-estimated on an expanding within-CEO window. Strictly look-ahead-free.	outputs/econometric/ceo_clarity_expanding/	thesis QtrExp variant of Dzieliński et al. (2021) Eqn 4

A.3 Financial / Market Variables

Name	Description / Formula	Source	Reference
`BGTAvg_Amihud`	BGT 25-day symmetric average Amihud illiquidity (H7e). *Formula:* Mean daily Amihud illiquidity over [-25, +25] trading days around call (51-day symmetric window, day 0 INCLUDED). F1D extension to BGT 2018.	.../variables/bgt_long_window_amihud.py	bgt2018 (window); F1D extension (symmetric average shape)
`BGTAvg_Spread`	BGT 25-day symmetric average bid-ask spread (H14e). *Formula:* Mean daily relative bid-ask spread over [-25, +25] trading days around call (51-day symmetric window, day 0 INCLUDED). Spread = (ASK - BID) / ((ASK + BID) / 2). F1D extension to BGT 2018.	.../variables/bgt_long_window_spread.py	bgt2018 (window); F1D extension (symmetric average shape)
`BGTDelta_Amihud`	BGT 25-day post-pre Amihud illiquidity change (H7d). *Formula:* Mean Amihud over post window [+1, +25] minus mean over pre window [-25, -1] trading days (day 0 EXCLUDED). F1D extension to BGT 2018.	.../variables/bgt_long_window_amihud.py	bgt2018 (window); F1D extension (delta shape)
`BGTDelta_Spread`	BGT 25-day post-pre bid-ask spread change (H14d). *Formula:* Mean spread over [+1, +25] minus mean over [-25, -1] trading days (day 0 EXCLUDED). F1D extension to BGT 2018.	.../variables/bgt_long_window_spread.py	bgt2018 (window); F1D extension (delta shape)
`BGTLevel_Amihud`	Compute BGT (2018) long-window Amihud illiquidity around earnings calls. *Formula:* CRSP via get_raw_daily_data (see module: src/f1d/shared/variables/bgt_long_window_amihud.py)	CRSPviaget_raw_daily_data	Bushee, Gow & Taylor (2018, JAR) 56(1):85-121
`BGTLevel_Spread`	Compute BGT (2018) long-window closing bid-ask spread around earnings calls. *Formula:* CRSP via get_raw_daily_data (closing BID/ASK) (see module: src/f1d/shared/variables/bgt_long_window_spread.py)	CRSPviaget_raw_daily_data	Bushee, Gow & Taylor (2018, JAR) [window]; Lee (2016, TAR) [formula]
`Capex`	Build Capex = capxy_annual / atq_lag from raw Compustat quarterly data. *Formula:* Compustat/capxy_Q4/atq (see module: src/f1d/shared/variables/capex_intensity.py)	Compustat/capxy_Q4/atq	Bates, Kahle & Stulz (2009, JF)
`CashFlowAt`	Build CashFlow = oancfy / atq from raw Compustat quarterly data. *Formula:* Compustat/oancfy/atq (see module: src/f1d/shared/variables/cash_flow.py)	Compustat/oancfy/atq	Bates, Kahle & Stulz (2009, JF)

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Name	Description / Formula	Source	Reference
`CashRatio`	Build CashRatio = cheq / atq from raw Compustat quarterly data. *Formula:* Compustat/cheq/atq (see module: src/f1d/shared/variables/cash_holdings.py)	Compustat/cheq/atq	Bates et al. (2009) (Section II, p.1991; BKS use annual che/at — we use quarterly cheq/atq for panel alignment)
`ChangDebtChoice`	ChangDebtChoice (description not in module docstring) *Formula:* see module: src/f1d/shared/variables/_compustat_engine.py	.../variables/_compustat_engine.py	cdh2006
`ChangExternalFunding`	ChangExternalFunding (description not in module docstring) *Formula:* see module: src/f1d/shared/variables/_compustat_engine.py	.../variables/_compustat_engine.py	cdh2006
`DailyVola`	Build Volatility from raw CRSP daily stock files via CRSPEngine. *Formula:* CRSP/RET (see module: src/f1d/shared/variables/volatility.py)	CRSP/RET	dwz2021
`DebtToCapital`	Build DebtToCapital = total debt / (equity + total debt). *Formula:* Compustat/dlcq,dlttq,seqq (see module: src/f1d/shared/variables/debt_to_capital.py)	Compustat/dlcq,dlttq,seqq	rz1995
`DivDummy`	Build DivDummy = (dvy ¿ 0).astype(float) from raw Compustat quarterly data. *Formula:* Compustat/dvy¿0 (see module: src/f1d/shared/variables/dividend_payer.py)	Compustat/dvy¿0	Bates et al. (2009) (Section IV, p.2000; dividend payout dummy based on common dividends)
`DivPayerQ`	Build DivPayerQ = (dvpsxq ¿ 0).astype(float) from raw Compustat quarterly data. *Formula:* Compustat/dvpsxq¿0 (quarterly ex-date) (see module: src/f1d/shared/variables/dividend_payer_quarterly.py)	Compustat/dvpsxq¿0	Hoberg & Prabhala (2009, RFS) — Compustat item 26 (dvpsx) analog
`EarnVol`	Builder for Earnings Volatility (earnings_volatility) variable. *Formula:* Compustat (see module: src/f1d/shared/variables/earnings_volatility.py)	Compustat	Duong, Do, and Do (2025) (Appendix B, p.30; 3-year rolling std of qoq asset-scaled earnings)
`EquityDelayCon`	Hoberg-Maksimovic (2015) equity-market financing constraint index (H22). *Formula:* Hoberg-Maksimovic (2015) firm-year equity-market financing constraint measure. Higher values = more constrained firm. Panel column is lowercase ‘equitydelaycon‘; CamelCase name is the spec convention. See docs/provenance/h22.md for merge details.	inputs/Hoberg_Maksimovic/	hm2015
`FirmMat`	Builder for Firm Maturity (firm_maturity) variable. *Formula:* Compustat (see module: src/f1d/shared/variables/firm_maturity.py)	Compustat	Biddle, Hilary, and Verdi (2009) (Appendix A, p.130; years since first CRSP appearance); alt Leary and Roberts (2010) (App.~A p.354)
`Leverage`	Build Leverage = (dlcq + dlttq) / atq (interest-bearing debt only). *Formula:* Compustat/dlcq,dlttq,atq (see module: src/f1d/shared/variables/book_lev.py)	Compustat/dlcq,dlttq,atq	bks2009, rz1995
`lnAssets`	Build Size = ln(atq) from raw Compustat quarterly data. *Formula:* Compustat/atq (see module: src/f1d/shared/variables/size.py)	Compustat/atq	dwz2021
`Loss`	Builder for Loss (H5 Control Variable). *Formula:* Compustat ibq ¡ 0 (see module: src/f1d/shared/variables/loss_dummy.py)	Compustatibq¡0	Biddle et al. (2009) (Appendix A, p.131; indicator for negative net income before extraordinary items)

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Name	Description / Formula	Source	Reference
`PayoutRatio_q`	Build PayoutRatio_q = (dvpspq × cshoq) / ibq from Compustat. *Formula:* Compustat/(dvpspq*cshoq)/ibq (quarterly) (see module: src/f1d/shared/variables/payout_ratio_quarterly.py)	Compustat/	dds2004
`PRisk`	Match quarterly PRisk onto each earnings call by (gvkey, calendar_quarter). *Formula:* Hassan et al. (2019) quarterly PRisk (see module: src/f1d/shared/variables/prisk_q.py)	Hassanetal.	hhlt2019
`RDSales`	Build RDSales = xrdq / atq from raw Compustat quarterly data. *Formula:* Compustat/xrdq,atq (see module: src/f1d/shared/variables/rd_intensity.py)	Compustat/xrdq,atq	bks2009, jjl2021
`RepurchaseIntensity`	Build RepurchaseIntensity = quarterly_prstkcy / lagged_atq from Compustat. *Formula:* Compustat/quarterly_prstkcy/lagged_atq (see module: src/f1d/shared/variables/repurchase_intensity.py)	Compustat/quarterly_prstkcy/lagged_atq	Thesis Advisor Suggestion
`ROA`	Build ROA = iby_annual / avg_assets from Compustat quarterly data. *Formula:* Compustat/iby,atq (see module: src/f1d/shared/variables/roa.py)	Compustat/iby,atq	Wang (2020), dwz2021
`SalesGrowth`	Build SalesGrowth from raw Compustat quarterly data (annual Q4 panel). *Formula:* Compustat/saley_Q4_YoY (see module: src/f1d/shared/variables/sales_growth.py)	Compustat/saley_Q4_YoY	Biddle et al. (2009) (Section 3.2, p.117; pct change in sales)
`sCFO`	Build sCFO = rolling 5-year std (min 3 yrs) of (oancfy/atq_{t-1}) per gvkey. *Formula:* Compustat/oancfy/atq_{t-1} rolling std (see module: src/f1d/shared/variables/ocf_volatility.py)	Compustat/oancfy/atq_{t-1}rollingstd	Biddle et al. (2009) (Appendix A, p.130; 5-year rolling std of CFO/avg assets)
`StockPrice`	Compute stock price at earnings call date (VECTORIZED). *Formula:* CRSP via get_raw_daily_data (see module: src/f1d/shared/variables/stock_price.py)	CRSPviaget_raw_daily_data	Amiram, Owens, and Rozenbaum (2016) (Appendix A, p.137; daily CRSP stock price; justification cites Stoll 1978)
`SurpDec`	Signed-quintile earnings surprise rank (-5 to +5) within calendar quarter. *Formula:* Signed quintile rank of percentage earnings surprise per DWZ 2021 Appendix Table A.1: raw surprise = (actual - consensus forecast earnings) / share price 5 trading days before announcement, x 100. Firms grouped into 5 bins of positive surprise (+5 largest through +1 smallest positive), 0 for zero surprises, and 5 bins of negative surprise (-1 smallest negative through -5 largest negative). DWZ footnote 32 notes this decile convention follows Hirshleifer, Lim & Teoh (2009) and DellaVigna & Pollet (2009).	.../variables/earnings_surprise.py	Dzieliński et al. (2021) (Appendix Table A.1, p.55; convention cites DellaVigna-Pollet 2009 + Hirshleifer-Lim-Teoh 2009)
`TobinsQ`	Build TobinsQ = (AT + cshoq*prccq - CEQ) / AT from raw Compustat quarterly data. *Formula:* Compustat/(atq+cshoq*prccq-ceqq)/atq (see module: src/f1d/shared/variables/tobins_q.py)	Compustat/	Bates, Kahle & Stulz (2009, JF)
`Turnover`	Compute share turnover at earnings call date (VECTORIZED). *Formula:* CRSP via get_raw_daily_data (see module: src/f1d/shared/variables/turnover.py)	CRSPviaget_raw_daily_data	Amihud (2002) (Section 2.1, p.33; alt rolling-window def in Chang, Dasgupta, and Hilary (2006) Appendix B p.3045)

A.4 Econometric / Indicator Variables

Name	Description / Formula	Source	Reference
BelowIG	Below-IG credit rating subsample indicator (H1.2). *Formula:* Binary = 1 if firm's S&P long-term issuer credit rating is below investment grade (BB+ and below) at call date; 0 otherwise. Rated firms only (Unrated firms get separate indicator).	runtime	thesis (H1.2 rating subsample)
HighCFvol	High cash-flow-volatility moderator (H1.3). *Formula:* Binary = 1 if firm-quarter Han-Qiu (2007) cash-flow-volatility (16-quarter rolling std of OCF / |mean| of OCF) is above the within-industry-year median; 0 otherwise.	runtime	Han and Qiu (2007); thesis (H1.3 split convention)
HighTSIMM	High product-market-similarity indicator (H1.1). *Formula:* Binary = 1 if firm-year TotalSimilarity (HP 2016 product market similarity) is above the within-year median; 0 otherwise.	runtime	hp2016 (TSIMM construction); thesis (H1.1 split convention)
Lagged_DV	Generic lagged-DV control (1Q lag of suite-specific DV). *Formula:* 1-quarter lag of the dependent variable, constructed at runtime per spec (DV varies per suite). Always included as base control to absorb persistence.	runtime	thesis convention (per feedback_lagged_dv.md)
Unrated	No-rating subsample indicator (H1.2). *Formula:* Binary = 1 if firm has no S&P rating at call date; 0 otherwise.	runtime	thesis (H1.2 rating subsample)
z_log_TotalSimilarity	z-scored log of HP 2016 TotalSimilarity (continuous moderator, H1.1). *Formula:* z-score of log(1 + TotalSimilarity) computed within year. TotalSimilarity = HP 2016 firm-year product market similarity score (sum of pairwise cosine similarities $\times$ 100).	runtime	hp2016

A.5 Runtime Derivatives (Lead/Lag/Centered/Interaction)

Name	Description / Formula	Source	Reference
AbsSurpDec	Absolute earnings surprise quintile magnitude. *Formula:* abs(SurpDec) — unsigned magnitude of signed-quintile earnings surprise rank, values 0-5.	runtime	thesis (magnitude control independent of sign)
BGTAvg_Amihud_lead1	1Q lead of BGTAvg_Amihud. *Formula:* 1-quarter lead of BGTAvg_Amihud	runtime	bgt2018 (window); F1D extension (symmetric average shape)
BGTAvg_Spread_lead1	1Q lead of BGTAvg_Spread. *Formula:* 1-quarter lead of BGTAvg_Spread	runtime	bgt2018 (window); F1D extension (symmetric average shape)
BGTDelta_Amihud_lead1	1Q lead of BGTDelta_Amihud. *Formula:* 1-quarter lead of BGTDelta_Amihud	runtime	bgt2018 (window); F1D extension (delta shape)
BGTDelta_Spread_lead1	1Q lead of BGTDelta_Spread. *Formula:* 1-quarter lead of BGTDelta_Spread	runtime	bgt2018 (window); F1D extension (delta shape)
BGTLevel_Amihud_lead1	1Q lead of BGTLevel_Amihud. *Formula:* 1-quarter lead of BGTLevel_Amihud	runtime	bgt2018

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Name	Description / Formula	Source	Reference
BGTLevel_Spread_lead1	1Q lead of BGTLevel_Spread. *Formula:* 1-quarter lead of BGTLevel_Spread	runtime	bgt2018, lee2016
Capex_lead	1Q lead of Capex. *Formula:* 1-quarter lead of Capex	runtime	Bates, Kahle & Stulz (2009, JF)
Capex_lead2	2Q lead of Capex. *Formula:* 2-quarter lead of Capex	runtime	Bates, Kahle & Stulz (2009, JF)
Capex_lead3	3Q lead of Capex. *Formula:* 3-quarter lead of Capex	runtime	Bates, Kahle & Stulz (2009, JF)
Capex_lead4	4Q lead of Capex. *Formula:* 4-quarter lead of Capex	runtime	Bates, Kahle & Stulz (2009, JF)
CashRatio_lead	1Q lead of CashRatio. *Formula:* 1-quarter lead of CashRatio	runtime	see base var
ChangDebtChoice_lead	1Q lead of ChangDebtChoice. *Formula:* 1-quarter lead of ChangDebtChoice	runtime	cdh2006
ChangExternalFunding_lead	1Q lead of ChangExternalFunding. *Formula:* 1-quarter lead of ChangExternalFunding	runtime	cdh2006
DebtToCapital_lead	1Q lead of DebtToCapital. *Formula:* 1-quarter lead of DebtToCapital	runtime	rz1995
DeltaILLIQ_lead1	1Q lead of DeltaILLIQ. *Formula:* 1-quarter lead of DeltaILLIQ	runtime	amihud02
DISP_lag	1Q lag of DISP. *Formula:* 1-quarter lag of DISP	runtime	Wang (2020)
DISP_lead	1Q lead of DISP. *Formula:* 1-quarter lead of DISP	runtime	Wang (2020)
DivPayerQ_lead1	1Q lead of DivPayerQ. *Formula:* 1-quarter lead of DivPayerQ	runtime	Hoberg & Prabhala (2009, RFS) – primary; Chetty & Saez (2005, QJE) – secondary (firm-quarter frequency precedent)
DSPREAD_lead1	1Q lead of DSPREAD. *Formula:* 1-quarter lead of DSPREAD	runtime	lee2016
EquityDelayCon_lead	1Q lead of EquityDelayCon. *Formula:* 1-quarter lead of EquityDelayCon	runtime	hm2015
Leverage_lead	1Q lead of Leverage. *Formula:* 1-quarter lead of Leverage	runtime	bks2009, rz1995

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Name	Description / Formula	Source	Reference
PayoutRatio_q_lead_qtr	1Q lead of PayoutRatio_q (qtr-aligned). *Formula:* 1-quarter lead of PayoutRatio_q (calendar-quarter aligned)	runtime	dds2004
PostCallAmihud	Panel-time post-call Amihud illiquidity level (H7b). *Formula:* Panel-time mean daily Amihud illiquidity over the post-call period (call quarter + N following quarters per H7b runner spec). Computed on panel structure with two-way clustering.	runtime	thesis (H7b panel-time post-call construction)
PostCallAmihud_lead1	1Q lead of PostCallAmihud. *Formula:* 1-quarter lead of PostCallAmihud	runtime	thesis (H7b panel-time post-call construction)
PostCallSpread	Panel-time post-call bid-ask spread level (H14b). *Formula:* Panel-time mean daily relative bid-ask spread over the post-call period (call quarter + N following quarters per H14b runner spec). Computed on panel structure with two-way clustering.	runtime	thesis (H14b panel-time post-call construction)
PostCallSpread_lead1	1Q lead of PostCallSpread. *Formula:* 1-quarter lead of PostCallSpread	runtime	thesis (H14b panel-time post-call construction)
PRisk_lag	1Q lag of PRisk. *Formula:* 1-quarter lag of PRisk	runtime	hhlt2019
PRisk_lag2	2Q lag of PRisk. *Formula:* 2-quarter lag of PRisk	runtime	hhlt2019
RDSales_lead	1Q lead of RDSales. *Formula:* 1-quarter lead of RDSales	runtime	bks2009, jjl2021
RepurchaseIntensity_lead_qtr	1Q lead of RepurchaseIntensity (qtr-aligned). *Formula:* 1-quarter lead of RepurchaseIntensity (calendar-quarter aligned)	runtime	Thesis Advisor Suggestion
ClarityCEO_c	Mean-centered ClarityCEO. *Formula:* ClarityCEO minus its sample mean (mean-centered for interaction interpretability)	runtime	thesis (interaction interpretability)
ClarityCEO_QtrExp_c	Mean-centered ClarityCEO_QtrExp. *Formula:* ClarityCEO_QtrExp minus its sample mean	runtime	thesis (interaction interpretability)
Post	DiD post-event-period indicator (Phase E sudden-death). *Formula:* Binary = 1 if call-quarter is on or after the death-event quarter for the treated firm (and on or after the matched event-quarter for the matched-control firm); 0 otherwise.	runtime	Ghafoor, Yousaf, and Li (2023)
Treated	DiD treated-firm indicator (Phase E sudden-death). *Formula:* Binary = 1 if firm experienced a CEO sudden-death event in 2002–2018; 0 if PSM-matched control firm.	runtime	Bennedsen, Pérez-González, and Wolfenzon (2020); Ghafoor et al. (2023)
Treated_x_Post	DiD ATT term (Phase E sudden-death). *Formula:* Treated $\times$ Post. The coefficient on this product is the average treatment effect on the treated.	runtime	Bertrand, Duflo, and Mullainathan (2004); Ghafoor et al. (2023)

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Name	Description / Formula	Source	Reference
UncPreCEO_c	Mean-centered UncPreCEO. *Formula:* UncPreCEO minus its sample mean (mean-centered for interaction interpretability)	runtime	thesis (interaction interpretability)
UncPreCEO_c_x_HighCFvol	Interaction: UncPreCEO_c × HighCFvol (H1.3). *Formula:* Product: UncPreCEO_c * HighCFvol	runtime	thesis (CFvol moderator interaction)
UncPreCEO_c_x_Unrated	Interaction: UncPreCEO_c × Unrated (HFC). *Formula:* Product: UncPreCEO_c * Unrated	runtime	thesis (HFC interaction)
UncResCEO_c	Mean-centered UncResCEO. *Formula:* UncResCEO minus its sample mean (mean-centered for interaction interpretability)	runtime	thesis (interaction interpretability)
UncResCEO_c_x_HighCFvol	Interaction: UncResCEO_c × HighCFvol (H1.3). *Formula:* Product: UncResCEO_c * HighCFvol	runtime	thesis (CFvol moderator interaction)
UncResCEO_c_x_Unrated	Interaction: UncResCEO_c × Unrated (HFC). *Formula:* Product: UncResCEO_c * Unrated	runtime	thesis (HFC interaction)
UncResCEO_QtrExp_c	Mean-centered UncResCEO_QtrExp. *Formula:* UncResCEO_QtrExp minus its sample mean	runtime	thesis (interaction interpretability)
UncResCEO_QtrExp_c_x_Unrated	Interaction: UncResCEO_QtrExp_c × Unrated (HFC QtrExp). *Formula:* Product: UncResCEO_QtrExp_c * Unrated	runtime	thesis (HFC QtrExp interaction)

A.6 Stage 9

Name	Description / Formula	Source	Reference
CCCL	SEC comment letter received (binary, forward-looking) *Formula:* 1 if SEC comment letter received in next quarter, 0 otherwise	inputs/EDGAR_CommentLetters	cdm2013
DeltaILLIQ	Change in Amihud illiquidity around earnings call *Formula:* PostCallILLIQ - PreCallILLIQ over [+1,+3] vs [-3,-1] days	inputs/CRSP_DSF	amihud02
DISP	Analyst forecast dispersion (Wang 2020) *Formula:* SD(latest analyst EPS forecasts in T-31..T-1 window) / prccq_prior	inputs/tr_ibes	Wang (2020)
DSPREAD	Change in closing bid-ask spread around earnings call *Formula:* mean(RelSpread[+1,+3]) - mean(RelSpread[-3,-1]) where RelSpread = 2*(ASK-BID)/(ASK+BID)	inputs/CRSP_DSF	lee2016
GEPU_log	Davis (2016, NBER WP 22740) Global Economic Policy Uncertainty (GEPU) index (log-transformed, current-$ GDP weights) *Formula:* log(GEPU_current) — GDP-weighted average of 16 country EPU indices, matched by calendar month of call start_date	inputs/EconomicPolicyUncertaintyIndex/Global	Davis (2016)

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Name	Description / Formula	Source	Reference
GPR_log	Caldara & Iacoviello (2022, AER) headline Geopolitical Risk index (log-transformed, 10-newspaper recent version) *Formula:* log(GPR) — recent headline Geopolitical Risk index, matched by calendar month of call start_date	inputs/matteoiacoviello/GPR_Global	ci2022
SEC_Letters_fwd	SEC comment letter count (forward-looking) *Formula:* Count of SEC-originated letters (UPLOAD type) in next calendar quarter	inputs/EDGAR_CommentLetters	cdm2013 (thesis extension to count)
TotalSimilarity	Hoberg-Phillips total product similarity *Formula:* Sum of pairwise cosine similarities from 10-K product descriptions	inputs/TNIC	hp2016
US_EPU_log	Baker, Bloom & Davis (2016, QJE) US News-Based Economic Policy Uncertainty index (log-transformed) *Formula:* log(News_Based_Policy_Uncert_Index) matched by calendar month of call start_date	inputs/EconomicPolicyUncertaintyIndex/US	bbd2016

Table 7: Summary Statistics

Variable	Unit	N	Mean	SD	Min	P25	Median	P75	Max
A. Independent Variables									
ClarityCEO	C	61,626	0.2143	0.2129	-0.7717	0.0952	0.2365	0.3629	0.8614
ClarityCEO_QtrExp	C	25,520	0.2011	0.2281	-1.0232	0.0713	0.2256	0.3589	0.8694
GEPU_log	C	88,205	4.6872	0.3672	3.8626	4.3797	4.7238	4.9666	5.5106
PRisk	C	85,995	99.6187	146.5189	0.0000	17.0778	53.3729	120.1973	1192.4308
PRisk_lag2	C	80,627	99.9614	147.2392	0.0000	17.1702	53.4446	120.2666	1192.4308
TotalSimilarity	F	22,403	3.4277	5.6522	1.0000	1.1246	1.5865	3.0903	80.1876
US_EPU_log	C	88,205	4.7123	0.3661	3.8018	4.4805	4.6891	4.9625	5.6478
UncPreCEO	C	71,727	0.6602	0.3917	0.0000	0.3810	0.6001	0.8723	2.2432
UncResCEO	C	44,900	-0.0000	0.3010	-1.5108	-0.1846	-0.0180	0.1650	1.7794
UncResCEO_QtrExp	C	25,520	-0.0057	0.3208	-1.6084	-0.2028	-0.0210	0.1707	1.7402
Unrated	C	79,487	0.5159	0.4997	0.0000	0.0000	1.0000	1.0000	1.0000
B. Dependent Variables									
BGTLevel_Spread	C	87,018	0.0017	0.0026	0.0001	0.0004	0.0009	0.0017	0.0164
BGTLevel_Spread_lead1	C	82,843	0.0016	0.0024	0.0001	0.0004	0.0009	0.0016	0.0164
CCCL	C	88,205	0.0032	0.0563	0.0000	0.0000	0.0000	0.0000	1.0000
CashRatio	C	87,990	0.1708	0.1806	0.0000	0.0368	0.1047	0.2435	0.9938
CashRatio_lead	C	82,236	0.1715	0.1748	0.0000	0.0407	0.1112	0.2447	0.9938
C. Firm Controls									
AbsSurpDec	C	61,544	2.9300	1.4255	0.0000	2.0000	3.0000	4.0000	5.0000
Capex	C	87,627	0.0487	0.0536	0.0000	0.0176	0.0324	0.0592	0.5203
CashFlowAt	C	87,670	0.1019	0.1147	-3.0324	0.0608	0.1032	0.1526	0.4874
DailyVola	C	82,599	36.7556	21.2437	9.4388	22.9709	31.2762	43.8148	211.9160
DivDummy	C	88,134	0.4755	0.4994	0.0000	0.0000	0.0000	1.0000	1.0000
EarnVol	C	87,440	0.0701	0.1907	0.0006	0.0163	0.0315	0.0695	19.5101
FirmMat	C	87,571	-0.0432	2.1826	-317.5714	-0.0078	0.2191	0.4195	0.9252
Leverage	C	87,994	0.2330	0.2240	0.0000	0.0470	0.2065	0.3455	3.9453
NegCall	C	86,858	0.9210	0.3024	0.0000	0.7038	0.8798	1.0943	2.3901
RDSales	C	87,647	0.1047	0.9342	0.0000	0.0000	0.0048	0.0636	39.7688
ROA	C	87,552	0.0386	0.1330	-3.9826	0.0146	0.0525	0.0925	0.5490
SalesGrowth	C	87,506	0.1173	0.3849	-1.0000	-0.0043	0.0745	0.1762	10.3051
StockPrice	C	87,188	36.8864	32.8119	1.6100	14.8975	28.2400	48.1700	192.0384
TobinsQ	C	87,363	2.1363	1.5197	0.3847	1.2681	1.6791	2.4362	35.6144
Turnover	C	87,188	0.0260	0.0299	0.0007	0.0083	0.0160	0.0310	0.1697
UncQue	C	81,031	1.4398	0.4831	0.0000	1.1216	1.4108	1.7268	3.5945
lnAssets	C	87,994	7.4710	1.6607	-0.5692	6.3008	7.3763	8.5436	12.4592
sCFO	C	83,662	0.0619	0.2233	0.0014	0.0236	0.0395	0.0664	32.4035

Notes: Summary statistics for variables used in the hypothesis tests. Sample period: 2002–2018. Anchor sample (H1, Main): 88,205 earnings-call observations across 1,884 firms. Unit column: C = call-level, F = firm-year. N varies by row because each variable is summarized on its anchor panel's Main sample (complete cases per variable); per-suite samples differ due to lags, control availability, and merges.

Table 8: H1 CEO 3-IV Decomp: ClarityCEO + UncResCEO (DWZ Full) + UncPreCEO

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	CashRatio						CashRatio_lead					
CEO Clarity (DWZ)	**-0.0046***	**-0.0071***	**-0.0058****	-0.0054	**-0.0056****	-0.0052	-0.0052	**-0.0170****	**-0.0068***	**-0.0162****	**-0.0069***	**-0.0161****
	(0.0033)	(0.0051)	(0.0033)	(0.0048)	(0.0033)	(0.0047)	(0.0052)	(0.0095)	(0.0051)	(0.0090)	(0.0051)	(0.0090)
CEO Residual Unc. (DWZ)	**0.0021****	**0.0019****	**0.0020****	**0.0016***	**0.0021****	**0.0016***	**0.0028****	**0.0023****	**0.0026****	**0.0020***	**0.0025****	**0.0019***
	(0.0010)	(0.0010)	(0.0010)	(0.0010)	(0.0010)	(0.0010)	(0.0014)	(0.0013)	(0.0014)	(0.0013)	(0.0014)	(0.0013)
CEO Pres Uncertainty	0.0002	0.0009	0.0003	0.0006	0.0004	0.0007	-0.0003	0.0008	-0.0005	0.0004	-0.0007	0.0002
	(0.0011)	(0.0011)	(0.0011)	(0.0011)	(0.0011)	(0.0011)	(0.0021)	(0.0017)	(0.0021)	(0.0017)	(0.0021)	(0.0017)
Leverage	**-0.0177*****	**-0.0355*****	**-0.0130*****	**-0.0218*****	**-0.0132*****	**-0.0216*****	**-0.0327*****	**-0.0279****	**-0.0274*****	-0.0159	**-0.0275*****	-0.0156
	(0.0048)	(0.0084)	(0.0036)	(0.0082)	(0.0037)	(0.0082)	(0.0120)	(0.0121)	(0.0104)	(0.0118)	(0.0105)	(0.0118)
lnAssets	**-0.0015*****	**-0.0177*****	**-0.0017*****	**-0.0175*****	**-0.0014*****	**-0.0170*****	**-0.0024****	**-0.0268*****	**-0.0020****	**-0.0265*****	**-0.0020****	**-0.0265*****
	(0.0006)	(0.0026)	(0.0005)	(0.0026)	(0.0005)	(0.0026)	(0.0010)	(0.0042)	(0.0009)	(0.0042)	(0.0010)	(0.0042)
TobinsQ	**0.0058*****	**0.0048*****	**0.0042*****	**0.0037*****	**0.0040*****	**0.0034*****	**0.0085*****	0.0025	**0.0056*****	0.0009	**0.0056*****	0.0010
	(0.0006)	(0.0009)	(0.0007)	(0.0010)	(0.0007)	(0.0010)	(0.0012)	(0.0015)	(0.0012)	(0.0015)	(0.0012)	(0.0016)
ROA	-0.0144	**0.0214****	**-0.0679*****	-0.0195	**-0.0630*****	-0.0171	**-0.0762*****	**-0.0352***	**-0.1427*****	**-0.0790*****	**-0.1378*****	**-0.0769*****
	(0.0093)	(0.0105)	(0.0113)	(0.0122)	(0.0114)	(0.0122)	(0.0161)	(0.0185)	(0.0173)	(0.0194)	(0.0174)	(0.0196)
Capex	**-0.1117*****	**-0.2995*****	**-0.1596*****	**-0.3074*****	**-0.1564*****	**-0.2993*****	**-0.1613*****	**-0.3281*****	**-0.2433*****	**-0.3415*****	**-0.2411*****	**-0.3430*****
	(0.0128)	(0.0234)	(0.0149)	(0.0236)	(0.0148)	(0.0234)	(0.0184)	(0.0344)	(0.0216)	(0.0345)	(0.0216)	(0.0346)
DivDummy	**-0.0041*****	-0.0028	**-0.0054*****	**-0.0041***	**-0.0049*****	-0.0036	**-0.0051****	-0.0068	**-0.0056****	**-0.0076***	**-0.0054****	**-0.0077***
	(0.0013)	(0.0023)	(0.0013)	(0.0022)	(0.0013)	(0.0022)	(0.0023)	(0.0045)	(0.0023)	(0.0044)	(0.0023)	(0.0044)
sCFO	**0.0066*****	-0.0022	**0.0079*****	-0.0008	**0.0074*****	-0.0009	**0.0137****	0.0027	**0.0137****	0.0035	**0.0134****	0.0034
	(0.0020)	(0.0014)	(0.0030)	(0.0014)	(0.0028)	(0.0014)	(0.0057)	(0.0025)	(0.0062)	(0.0027)	(0.0061)	(0.0027)
Lagged_DV	**0.8558*****	**0.6146*****	**0.8612*****	**0.6322*****	**0.8617*****	**0.6337*****	**0.7214*****	**0.2209*****	**0.7227*****	**0.2377*****	**0.7231*****	**0.2385*****
	(0.0076)	(0.0133)	(0.0075)	(0.0130)	(0.0075)	(0.0130)	(0.0138)	(0.0201)	(0.0133)	(0.0201)	(0.0133)	(0.0201)
SalesGrowth			**-0.0178*****	**-0.0274*****	**-0.0169*****	**-0.0266*****			**-0.0156****	**-0.0171*****	**-0.0144****	**-0.0163*****
			(0.0045)	(0.0052)	(0.0044)	(0.0052)			(0.0061)	(0.0050)	(0.0060)	(0.0050)
RDSales			**0.0062*****	**-0.0032*****	**0.0062*****	**-0.0031****			**0.0137*****	-0.0014	**0.0138*****	-0.0012
			(0.0017)	(0.0012)	(0.0017)	(0.0012)			(0.0024)	(0.0019)	(0.0024)	(0.0019)
CashFlowAt			**0.1292*****	**0.1589*****	**0.1278*****	**0.1576*****			**0.1982*****	**0.1687*****	**0.1944*****	**0.1660*****
			(0.0162)	(0.0173)	(0.0162)	(0.0173)			(0.0236)	(0.0249)	(0.0237)	(0.0250)
DailyVola			0.0000	**-0.0001*****	**0.0001*****	**-0.0001***			**0.0003*****	**0.0001*****	**0.0003*****	0.0001
			(0.0000)	(0.0000)	(0.0000)	(0.0000)			(0.0000)	(0.0000)	(0.0001)	(0.0001)
Extended Controls			Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes
Industry FE	Yes		Yes		Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes		Yes		Yes
Year FE	Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes		
Year-Quarter FE					Yes	Yes					Yes	Yes
N	43,333	43,333	43,316	43,316	43,316	43,316	41,122	41,122	41,108	41,108	41,108	41,108
R^2	0.825	0.431	0.829	0.449	0.830	0.449	0.659	0.092	0.667	0.108	0.668	0.108
Adj. R^2	0.825	0.412	0.829	0.430	0.830	0.430	0.658	0.062	0.667	0.078	0.667	0.077

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). R^2 includes absorbed fixed effects (not within-R^2).

Table 9: H1 CEO 3-IV Decomp: ClarityCEO + UncResCEO (DWZ Qtr-Exp, no-look-ahead) + UncPreCEO

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	CashRatio						CashRatio_lead					
CEO Clarity (DWZ Qtr-Exp)	**-0.0083*****	**-0.0148*****	**-0.0093*****	**-0.0129*****	**-0.0092*****	**-0.0126*****	-0.0024	**-0.0215****	-0.0041	**-0.0210****	-0.0042	**-0.0210****
	(0.0033)	(0.0047)	(0.0033)	(0.0045)	(0.0033)	(0.0045)	(0.0053)	(0.0097)	(0.0053)	(0.0094)	(0.0053)	(0.0095)
CEO Residual Unc. (DWZ Qtr-Exp)	**0.0018***	**0.0020****	**0.0016***	**0.0018***	**0.0016***	**0.0017***	**0.0046*****	**0.0036****	**0.0044****	**0.0033****	**0.0044****	**0.0033****
	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0019)	(0.0017)	(0.0019)	(0.0017)	(0.0019)	(0.0017)
CEO Pres Uncertainty	0.0001	0.0006	-0.0001	0.0000	-0.0001	0.0001	0.0002	0.0016	-0.0003	0.0011	-0.0004	0.0011
	(0.0013)	(0.0013)	(0.0013)	(0.0013)	(0.0013)	(0.0013)	(0.0024)	(0.0021)	(0.0024)	(0.0021)	(0.0024)	(0.0021)
Leverage	**-0.0131*****	**-0.0324*****	**-0.0089*****	**-0.0199****	**-0.0087*****	**-0.0190****	**-0.0293****	**-0.0295****	**-0.0241****	-0.0181	**-0.0244****	-0.0179
	(0.0040)	(0.0086)	(0.0031)	(0.0085)	(0.0031)	(0.0085)	(0.0122)	(0.0139)	(0.0108)	(0.0136)	(0.0109)	(0.0136)
lnAssets	-0.0009	**-0.0169*****	**-0.0011***	**-0.0169*****	**-0.0011***	**-0.0167*****	**-0.0018***	**-0.0285*****	-0.0016	**-0.0286*****	-0.0015	**-0.0287*****
	(0.0006)	(0.0028)	(0.0006)	(0.0027)	(0.0006)	(0.0027)	(0.0011)	(0.0049)	(0.0010)	(0.0049)	(0.0011)	(0.0050)
TobinsQ	**0.0052*****	**0.0039*****	**0.0034*****	**0.0028****	**0.0033*****	**0.0025****	**0.0079*****	0.0011	**0.0046*****	-0.0009	**0.0046*****	-0.0009
	(0.0007)	(0.0011)	(0.0009)	(0.0012)	(0.0009)	(0.0012)	(0.0012)	(0.0017)	(0.0014)	(0.0017)	(0.0014)	(0.0018)
ROA	-0.0107	**0.0230***	**-0.0510*****	-0.0135	**-0.0498*****	-0.0129	**-0.0621*****	-0.0188	**-0.1280*****	**-0.0721*****	**-0.1270*****	**-0.0731*****
	(0.0109)	(0.0124)	(0.0131)	(0.0144)	(0.0132)	(0.0144)	(0.0192)	(0.0227)	(0.0202)	(0.0237)	(0.0204)	(0.0240)
Capex	**-0.0967*****	**-0.2472*****	**-0.1415*****	**-0.2628*****	**-0.1414*****	**-0.2605*****	**-0.1613*****	**-0.2816*****	**-0.2420*****	**-0.3002*****	**-0.2423*****	**-0.2996*****
	(0.0150)	(0.0253)	(0.0169)	(0.0258)	(0.0168)	(0.0257)	(0.0214)	(0.0372)	(0.0249)	(0.0375)	(0.0249)	(0.0375)
DivDummy	**-0.0037****	-0.0021	**-0.0051*****	-0.0034	**-0.0050*****	-0.0033	**-0.0042***	-0.0080	**-0.0048****	**-0.0090***	**-0.0047***	**-0.0091***
	(0.0015)	(0.0026)	(0.0015)	(0.0025)	(0.0015)	(0.0025)	(0.0025)	(0.0054)	(0.0025)	(0.0053)	(0.0025)	(0.0053)
sCFO	**0.0090*****	-0.0008	**0.0100*****	0.0004	**0.0099*****	0.0004	**0.0147***	0.0031	**0.0143***	0.0038	**0.0142***	0.0037
	(0.0024)	(0.0017)	(0.0031)	(0.0014)	(0.0030)	(0.0015)	(0.0081)	(0.0032)	(0.0082)	(0.0032)	(0.0081)	(0.0032)
Lagged_DV	**0.8718*****	**0.6433*****	**0.8734*****	**0.6576*****	**0.8738*****	**0.6579*****	**0.7424*****	**0.2158*****	**0.7404*****	**0.2320*****	**0.7402*****	**0.2317*****
	(0.0091)	(0.0159)	(0.0091)	(0.0156)	(0.0091)	(0.0156)	(0.0150)	(0.0242)	(0.0148)	(0.0242)	(0.0148)	(0.0242)
SalesGrowth			**-0.0165*****	**-0.0257*****	**-0.0165*****	**-0.0256*****			-0.0106	**-0.0137****	-0.0107	**-0.0136****
			(0.0054)	(0.0068)	(0.0054)	(0.0068)			(0.0070)	(0.0064)	(0.0070)	(0.0064)
RDSales			**0.0091*****	-0.0004	**0.0091*****	-0.0003			**0.0160*****	0.0005	**0.0159*****	0.0005
			(0.0025)	(0.0017)	(0.0025)	(0.0017)			(0.0038)	(0.0021)	(0.0038)	(0.0021)
CashFlowAt			**0.1182*****	**0.1536*****	**0.1189*****	**0.1544*****			**0.1977*****	**0.1883*****	**0.1974*****	**0.1880*****
			(0.0159)	(0.0175)	(0.0159)	(0.0175)			(0.0275)	(0.0300)	(0.0275)	(0.0300)
DailyVola			0.0000	**-0.0001****	0.0000	**-0.0001****			**0.0002*****	0.0000	**0.0003*****	0.0000
			(0.0000)	(0.0000)	(0.0000)	(0.0000)			(0.0001)	(0.0000)	(0.0001)	(0.0001)
Extended Controls			Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes
Industry FE	Yes		Yes		Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes		Yes		Yes
Year FE	Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes		
Year-Quarter FE					Yes	Yes					Yes	Yes
N	24,868	24,868	24,855	24,855	24,855	24,855	23,127	23,127	23,120	23,120	23,120	23,120
R^2	0.839	0.457	0.842	0.473	0.842	0.473	0.675	0.087	0.683	0.105	0.683	0.105
Adj. R^2	0.839	0.430	0.841	0.446	0.842	0.446	0.674	0.041	0.682	0.059	0.682	0.058

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). R^2 includes absorbed fixed effects (not within-R^2).

Table 10: H1.2 CEO 3-IV Decomp HFC: ClarityCEO + UncResCEO (DWZ Full) + UncPreCEO × Unrated Constraint

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	CashRatio				CashRatio_lead			
CEO Clarity (DWZ, c)	**-0.0062****	-0.0043	**-0.0059****	-0.0042	-0.0052	**-0.0157***	-0.0052	**-0.0155***
	(0.0034)	(0.0052)	(0.0034)	(0.0052)	(0.0053)	(0.0096)	(0.0053)	(0.0096)
CEO Residual Unc. (DWZ, c)	**0.0020****	**0.0018****	**0.0022****	**0.0019****	**0.0027****	**0.0022***	**0.0026****	**0.0021***
	(0.0011)	(0.0010)	(0.0011)	(0.0010)	(0.0015)	(0.0013)	(0.0015)	(0.0013)
CEO Pres Unc. (c)	0.0002	0.0005	0.0003	0.0007	-0.0007	-0.0001	-0.0008	-0.0002
	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0022)	(0.0018)	(0.0022)	(0.0018)
Unrated	**0.0038*****	-0.0027	-0.0010	-0.0026	**0.0065*****	**-0.0093***	-0.0014	**-0.0095***
	(0.0012)	(0.0035)	(0.0016)	(0.0035)	(0.0022)	(0.0054)	(0.0029)	(0.0054)
UncRes × Unrated	0.0011	-0.0001	0.0012	-0.0000	**0.0057****	0.0033	**0.0057****	0.0033
	(0.0021)	(0.0020)	(0.0021)	(0.0020)	(0.0029)	(0.0026)	(0.0029)	(0.0026)
UncPre × Unrated	0.0024	0.0013	**0.0029***	0.0014	0.0005	-0.0026	0.0010	-0.0027
	(0.0022)	(0.0025)	(0.0023)	(0.0025)	(0.0040)	(0.0038)	(0.0040)	(0.0038)
Leverage	**-0.0117*****	**-0.0216****	**-0.0141*****	**-0.0214****	**-0.0237****	-0.0107	**-0.0274****	-0.0104
	(0.0033)	(0.0098)	(0.0041)	(0.0099)	(0.0103)	(0.0131)	(0.0117)	(0.0131)
lnAssets	**0.0005***	**-0.0159*****	**-0.0019*****	**-0.0155*****	**0.0018*****	**-0.0273*****	**-0.0025****	**-0.0275*****
	(0.0003)	(0.0027)	(0.0007)	(0.0027)	(0.0005)	(0.0044)	(0.0011)	(0.0044)
TobinsQ	**0.0047*****	**0.0045*****	**0.0041*****	**0.0040*****	**0.0059*****	0.0009	**0.0052*****	0.0010
	(0.0008)	(0.0011)	(0.0008)	(0.0011)	(0.0013)	(0.0017)	(0.0013)	(0.0017)
ROA	**-0.0715*****	-0.0157	**-0.0651*****	-0.0133	**-0.1452*****	**-0.0819*****	**-0.1378*****	**-0.0800*****
	(0.0122)	(0.0122)	(0.0122)	(0.0122)	(0.0182)	(0.0200)	(0.0183)	(0.0202)
Capex	**-0.1610*****	**-0.3087*****	**-0.1647*****	**-0.3005*****	**-0.2338*****	**-0.3399*****	**-0.2426*****	**-0.3419*****
	(0.0158)	(0.0254)	(0.0158)	(0.0251)	(0.0226)	(0.0360)	(0.0227)	(0.0361)
DivDummy	**-0.0058*****	**-0.0043***	**-0.0054*****	-0.0038	**-0.0054****	-0.0065	**-0.0055****	-0.0066
	(0.0013)	(0.0023)	(0.0013)	(0.0023)	(0.0023)	(0.0044)	(0.0023)	(0.0044)
sCFO	**0.0081*****	-0.0003	**0.0072*****	-0.0005	**0.0135****	0.0043	**0.0125****	0.0042
	(0.0031)	(0.0016)	(0.0026)	(0.0016)	(0.0062)	(0.0029)	(0.0055)	(0.0029)
SalesGrowth	**-0.0189*****	**-0.0297*****	**-0.0176*****	**-0.0287*****	**-0.0158****	**-0.0156*****	**-0.0143****	**-0.0148*****
	(0.0045)	(0.0048)	(0.0044)	(0.0048)	(0.0062)	(0.0046)	(0.0062)	(0.0047)
RDSales	**0.0089*****	-0.0018	**0.0090*****	-0.0016	**0.0152*****	-0.0004	**0.0155*****	-0.0003
	(0.0020)	(0.0015)	(0.0020)	(0.0015)	(0.0032)	(0.0024)	(0.0032)	(0.0024)
CashFlowAt	**0.1415*****	**0.1741*****	**0.1368*****	**0.1729*****	**0.2109*****	**0.1796*****	**0.2012*****	**0.1767*****
	(0.0166)	(0.0180)	(0.0165)	(0.0180)	(0.0240)	(0.0257)	(0.0241)	(0.0259)
DailyVola	**0.0001****	**-0.0001*****	**0.0001****	-0.0001	**0.0004*****	**0.0001*****	**0.0003*****	0.0001
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0000)	(0.0001)	(0.0001)
Lagged_DV	**0.8605*****	**0.6188*****	**0.8571*****	**0.6204*****	**0.7310*****	**0.2231*****	**0.7250*****	**0.2242*****
	(0.0079)	(0.0133)	(0.0080)	(0.0134)	(0.0135)	(0.0193)	(0.0138)	(0.0193)
Industry FE	Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes
Year FE	Yes	Yes			Yes	Yes		
Year-Quarter FE			Yes	Yes			Yes	Yes
N	38,737	38,737	38,737	38,737	38,092	38,092	38,092	38,092
R^2	0.827	0.432	0.829	0.432	0.667	0.100	0.669	0.100
Adj. R^2	0.827	0.412	0.828	0.411	0.667	0.069	0.668	0.067

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Fiscal years 2002-2016. DWZ Eq.5 decomposition: Clarity = -CEO FE; UncRes = call-level residual. R^2 includes absorbed fixed effects (not within-R^2).

Table 11: H1.2 CEO 3-IV Decomp HFC: ClarityCEO + UncResCEO (DWZ Qtr-Exp, no-look-ahead) + UncPreCEO × Unrated Constraint

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	CashRatio				CashRatio_lead			
CEO Clarity (DWZ Qtr-Exp, c)	**-0.0114*****	**-0.0132*****	**-0.0104*****	**-0.0127*****	-0.0037	**-0.0186****	-0.0019	**-0.0187****
	(0.0035)	(0.0052)	(0.0034)	(0.0053)	(0.0054)	(0.0103)	(0.0054)	(0.0103)
CEO Residual Unc. (DWZ Qtr-Exp, c)	**0.0021***	**0.0024****	**0.0021***	**0.0022****	**0.0044****	**0.0036****	**0.0044****	**0.0036****
	(0.0013)	(0.0013)	(0.0013)	(0.0013)	(0.0020)	(0.0017)	(0.0020)	(0.0017)
CEO Pres Unc. (c)	0.0000	-0.0005	0.0000	-0.0004	0.0003	0.0009	0.0000	0.0009
	(0.0013)	(0.0013)	(0.0013)	(0.0013)	(0.0025)	(0.0022)	(0.0025)	(0.0022)
Unrated	0.0021	-0.0043	-0.0012	-0.0043	**0.0051****	-0.0073	-0.0019	-0.0074
	(0.0014)	(0.0043)	(0.0018)	(0.0043)	(0.0024)	(0.0066)	(0.0032)	(0.0066)
UncRes (Qtr-Exp) × Unrated	0.0019	0.0013	0.0019	0.0013	**0.0050***	0.0023	**0.0050***	0.0022
	(0.0025)	(0.0023)	(0.0025)	(0.0023)	(0.0038)	(0.0033)	(0.0038)	(0.0033)
UncPre × Unrated	0.0014	-0.0014	0.0017	-0.0014	-0.0001	-0.0015	0.0004	-0.0014
	(0.0026)	(0.0029)	(0.0026)	(0.0029)	(0.0046)	(0.0046)	(0.0046)	(0.0046)
Leverage	**-0.0084*****	-0.0170	**-0.0094*****	-0.0160	**-0.0210***	-0.0110	**-0.0240****	-0.0107
	(0.0031)	(0.0104)	(0.0034)	(0.0104)	(0.0108)	(0.0154)	(0.0121)	(0.0154)
lnAssets	0.0003	**-0.0157*****	**-0.0016***	**-0.0157*****	**0.0016*****	**-0.0300*****	**-0.0021***	**-0.0302*****
	(0.0003)	(0.0031)	(0.0008)	(0.0031)	(0.0006)	(0.0052)	(0.0013)	(0.0053)
TobinsQ	**0.0039*****	**0.0030****	**0.0034*****	**0.0025***	**0.0049*****	-0.0011	**0.0044*****	-0.0011
	(0.0010)	(0.0015)	(0.0010)	(0.0014)	(0.0014)	(0.0019)	(0.0015)	(0.0020)
ROA	**-0.0554*****	-0.0071	**-0.0535*****	-0.0065	**-0.1336*****	**-0.0761*****	**-0.1305*****	**-0.0773*****
	(0.0139)	(0.0145)	(0.0140)	(0.0146)	(0.0211)	(0.0249)	(0.0213)	(0.0252)
Capex	**-0.1396*****	**-0.2564*****	**-0.1439*****	**-0.2536*****	**-0.2276*****	**-0.2870*****	**-0.2378*****	**-0.2865*****
	(0.0175)	(0.0276)	(0.0176)	(0.0275)	(0.0263)	(0.0384)	(0.0262)	(0.0384)
DivDummy	**-0.0055*****	-0.0041	**-0.0056*****	-0.0041	**-0.0044***	-0.0076	**-0.0046***	-0.0077
	(0.0016)	(0.0027)	(0.0016)	(0.0026)	(0.0025)	(0.0053)	(0.0026)	(0.0053)
sCFO	**0.0099*****	0.0017	**0.0094*****	0.0017	**0.0123***	0.0041	**0.0116***	0.0040
	(0.0030)	(0.0016)	(0.0028)	(0.0016)	(0.0069)	(0.0032)	(0.0063)	(0.0032)
SalesGrowth	**-0.0128****	**-0.0267*****	**-0.0128****	**-0.0265*****	-0.0081	**-0.0127****	-0.0083	**-0.0127****
	(0.0051)	(0.0056)	(0.0051)	(0.0056)	(0.0074)	(0.0063)	(0.0075)	(0.0063)
RDSales	**0.0086*****	-0.0004	**0.0086*****	-0.0003	**0.0154*****	0.0010	**0.0154*****	0.0011
	(0.0022)	(0.0017)	(0.0022)	(0.0017)	(0.0035)	(0.0022)	(0.0035)	(0.0022)
CashFlowAt	**0.1220*****	**0.1745*****	**0.1204*****	**0.1754*****	**0.2021*****	**0.1992*****	**0.1968*****	**0.1990*****
	(0.0169)	(0.0188)	(0.0169)	(0.0189)	(0.0288)	(0.0317)	(0.0288)	(0.0317)
DailyVola	0.0000	**-0.0001****	-0.0000	**-0.0001****	**0.0003*****	0.0000	**0.0002*****	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0000)	(0.0001)	(0.0001)
Lagged_DV	**0.8737*****	**0.6448*****	**0.8713*****	**0.6451*****	**0.7471*****	**0.2124*****	**0.7408*****	**0.2122*****
	(0.0096)	(0.0166)	(0.0096)	(0.0166)	(0.0152)	(0.0230)	(0.0154)	(0.0229)
Industry FE	Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes
Year FE	Yes	Yes			Yes	Yes		
Year-Quarter FE			Yes	Yes			Yes	Yes
N	21,824	21,824	21,824	21,824	21,371	21,371	21,371	21,371
R^2	0.841	0.453	0.841	0.453	0.679	0.094	0.680	0.094
Adj. R^2	0.841	0.423	0.841	0.423	0.678	0.046	0.679	0.044

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Fiscal years 2002-2016. DWZ Eq.5 decomposition under quarterly-expanding window (no-look-ahead). R^2 includes absorbed fixed effects (not within-R^2).

Table 12: H1.3 CEO 3-IV Decomp CFvol: ClarityCEO + UncResCEO (DWZ Full) + UncPreCEO × HighCFvol (Han-Qiu 16q CV)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	CashRatio				CashRatio_lead			
CEO Clarity (DWZ, c)	**-0.0088*****	-0.0059	**-0.0071****	-0.0058	**-0.0096****	**-0.0135***	**-0.0078***	**-0.0134***
	(0.0034)	(0.0047)	(0.0033)	(0.0047)	(0.0052)	(0.0091)	(0.0052)	(0.0091)
CEO Residual Unc. (DWZ, c)	**0.0021****	**0.0017***	**0.0022****	**0.0017****	**0.0025****	0.0016	**0.0024***	0.0015
	(0.0011)	(0.0010)	(0.0011)	(0.0010)	(0.0015)	(0.0013)	(0.0015)	(0.0014)
CEO Pres Unc. (c)	0.0004	0.0006	0.0001	0.0007	-0.0002	0.0004	-0.0008	0.0002
	(0.0011)	(0.0011)	(0.0012)	(0.0012)	(0.0021)	(0.0018)	(0.0022)	(0.0018)
HighCFvol	-0.0005	-0.0022	**-0.0028****	-0.0023	**0.0075*****	0.0021	**0.0049****	0.0021
	(0.0012)	(0.0016)	(0.0013)	(0.0016)	(0.0019)	(0.0025)	(0.0020)	(0.0025)
UncRes × HighCFvol	0.0004	0.0004	-0.0000	0.0002	**0.0047***	0.0028	**0.0048***	0.0030
	(0.0020)	(0.0019)	(0.0020)	(0.0019)	(0.0030)	(0.0026)	(0.0029)	(0.0026)
UncPre × HighCFvol	**0.0048****	0.0002	**0.0049****	0.0002	**0.0099*****	0.0040	**0.0101*****	0.0041
	(0.0022)	(0.0024)	(0.0022)	(0.0024)	(0.0039)	(0.0040)	(0.0039)	(0.0040)
Leverage	**-0.0127*****	**-0.0191****	**-0.0130*****	**-0.0189****	**-0.0273*****	-0.0116	**-0.0274*****	-0.0112
	(0.0034)	(0.0082)	(0.0036)	(0.0082)	(0.0094)	(0.0121)	(0.0097)	(0.0121)
lnAssets	0.0004	**-0.0174*****	**-0.0015*****	**-0.0169*****	**0.0014*****	**-0.0236*****	-0.0013	**-0.0235*****
	(0.0003)	(0.0026)	(0.0006)	(0.0026)	(0.0005)	(0.0043)	(0.0010)	(0.0044)
TobinsQ	**0.0044*****	**0.0037*****	**0.0035*****	**0.0034*****	**0.0066*****	0.0016	**0.0056*****	0.0017
	(0.0008)	(0.0011)	(0.0008)	(0.0011)	(0.0013)	(0.0017)	(0.0014)	(0.0018)
ROA	**-0.0641*****	-0.0148	**-0.0563*****	-0.0121	**-0.1405*****	**-0.0736*****	**-0.1330*****	**-0.0717*****
	(0.0117)	(0.0123)	(0.0116)	(0.0123)	(0.0185)	(0.0205)	(0.0186)	(0.0206)
Capex	**-0.1692*****	**-0.3164*****	**-0.1779*****	**-0.3085*****	**-0.2443*****	**-0.3482*****	**-0.2565*****	**-0.3496*****
	(0.0158)	(0.0251)	(0.0160)	(0.0249)	(0.0229)	(0.0363)	(0.0232)	(0.0364)
DivDummy	**-0.0057*****	**-0.0051****	**-0.0054*****	**-0.0046****	**-0.0048****	**-0.0085***	**-0.0049****	**-0.0086***
	(0.0013)	(0.0023)	(0.0013)	(0.0023)	(0.0023)	(0.0046)	(0.0023)	(0.0046)
sCFO	**0.0087****	0.0004	**0.0078****	0.0003	**0.0128***	0.0041	**0.0121***	0.0040
	(0.0043)	(0.0017)	(0.0037)	(0.0017)	(0.0071)	(0.0031)	(0.0066)	(0.0030)
SalesGrowth	**-0.0203*****	**-0.0291*****	**-0.0192*****	**-0.0283*****	**-0.0187*****	**-0.0188*****	**-0.0172****	**-0.0181*****
	(0.0050)	(0.0057)	(0.0049)	(0.0057)	(0.0068)	(0.0055)	(0.0067)	(0.0056)
RDSales	**0.0063*****	**-0.0031****	**0.0063*****	**-0.0029****	**0.0153*****	-0.0015	**0.0154*****	-0.0014
	(0.0022)	(0.0015)	(0.0022)	(0.0015)	(0.0037)	(0.0029)	(0.0036)	(0.0029)
CashFlowAt	**0.1476*****	**0.1713*****	**0.1382*****	**0.1698*****	**0.2228*****	**0.1725*****	**0.2093*****	**0.1698*****
	(0.0170)	(0.0179)	(0.0171)	(0.0179)	(0.0255)	(0.0262)	(0.0258)	(0.0264)
DailyVola	**0.0001*****	**-0.0001*****	**0.0001*****	-0.0001	**0.0004*****	**0.0001*****	**0.0003*****	0.0001
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0000)	(0.0001)	(0.0001)
Lagged_DV	**0.8664*****	**0.6307*****	**0.8619*****	**0.6322*****	**0.7262*****	**0.2312*****	**0.7200*****	**0.2320*****
	(0.0077)	(0.0133)	(0.0078)	(0.0133)	(0.0139)	(0.0205)	(0.0142)	(0.0205)
Industry FE	Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes
Year FE	Yes	Yes			Yes	Yes		
Year-Quarter FE			Yes	Yes			Yes	Yes
N	40,795	40,795	40,795	40,795	38,671	38,671	38,671	38,671
R^2	0.825	0.447	0.826	0.447	0.657	0.103	0.659	0.102
Adj. R^2	0.824	0.429	0.826	0.428	0.657	0.072	0.658	0.071

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Fiscal years 2002-2018. Full panel coverage via Compustat Quarterly OCF Extended (1990-2002 backfill); Han-Qiu 16-trailing-quarter CFvol available from 1994-Q1 onwards in source data. R^2 includes absorbed fixed effects (not within-R^2).

Table 13: H11: Political Risk and Language Uncertainty

	(1)	(2)	(3)	(4)
	Industry FE		Firm FE	
	UncResCEO	UncPreCEO	UncResCEO	UncPreCEO
Political Risk$_t$	**0.0001*****	**0.0003*****	**0.0001*****	**0.0002*****
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
UncQue	-0.0022	**0.0204*****	-0.0014	**0.0081*****
	(0.0032)	(0.0045)	(0.0037)	(0.0029)
NegCall	-0.0025	**0.1874*****	-0.0036	**0.1477*****
	(0.0051)	(0.0141)	(0.0073)	(0.0082)
lnAssets	-0.0002	**-0.0296*****	-0.0054	0.0048
	(0.0004)	(0.0039)	(0.0058)	(0.0078)
TobinsQ	-0.0005	-0.0007	-0.0008	0.0001
	(0.0009)	(0.0036)	(0.0022)	(0.0023)
ROA	**0.0291****	0.0477	**0.0540****	0.0346
	(0.0145)	(0.0318)	(0.0264)	(0.0230)
CashRatio	0.0019	-0.0354	0.0206	0.0265
	(0.0077)	(0.0357)	(0.0240)	(0.0287)
DivDummy	-0.0031	-0.0139	-0.0108	**-0.0229****
	(0.0021)	(0.0118)	(0.0074)	(0.0103)
FirmMat	0.0012	**0.0021****	0.0027	0.0006
	(0.0013)	(0.0010)	(0.0041)	(0.0007)
EarnVol	0.0078	-0.0072	0.0056	-0.0104
	(0.0077)	(0.0157)	(0.0117)	(0.0132)
UncPreCEO	-0.0051		-0.0054	
	(0.0034)		(0.0054)	
Industry FE	Yes	Yes		
Firm FE			Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	44,497	65,760	44,497	65,760
R^2	0.003	0.043	0.003	0.023
Adj. R^2	0.002	0.043	-0.030	-0.004

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Firms with fewer than 5 calls are excluded. R^2 includes absorbed fixed effects (not within-R^2).

Table 14: H11-Lag2: Political Risk (2-Qtr Lag) and Language Uncertainty

	(1)	(2)	(3)	(4)
	Industry FE		Firm FE	
	UncResCEO	UncPreCEO	UncResCEO	UncPreCEO
$PRisk_{t-2}$	0.0000	**0.0001*****	0.0000	**0.0000*****
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
UncQue	0.0001	**0.0242*****	0.0010	**0.0099*****
	(0.0033)	(0.0047)	(0.0038)	(0.0031)
NegCall	0.0020	**0.2019*****	0.0019	**0.1609*****
	(0.0052)	(0.0145)	(0.0074)	(0.0084)
lnAssets	-0.0001	**-0.0298*****	-0.0035	0.0047
	(0.0005)	(0.0040)	(0.0062)	(0.0078)
TobinsQ	-0.0005	-0.0002	-0.0011	-0.0002
	(0.0010)	(0.0037)	(0.0023)	(0.0023)
ROA	**0.0293***	0.0518	**0.0598****	**0.0464***
	(0.0152)	(0.0331)	(0.0272)	(0.0247)
CashRatio	0.0064	-0.0400	0.0170	0.0233
	(0.0076)	(0.0368)	(0.0242)	(0.0292)
DivDummy	-0.0031	-0.0142	**-0.0133***	**-0.0207***
	(0.0022)	(0.0121)	(0.0076)	(0.0106)
FirmMat	0.0013	0.0021	0.0025	-0.0007
	(0.0013)	(0.0015)	(0.0041)	(0.0016)
EarnVol	0.0096	-0.0078	0.0089	-0.0154
	(0.0063)	(0.0159)	(0.0100)	(0.0158)
UncPreCEO	-0.0001		-0.0003	
	(0.0035)		(0.0056)	
Industry FE	Yes	Yes		
Firm FE			Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	43,037	63,022	43,037	63,022
R^2	0.000	0.035	0.000	0.015
Adj. R^2	-0.001	0.034	-0.033	-0.013

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Firms with fewer than 5 calls are excluded. R^2 includes absorbed fixed effects (not within-R^2).

Table 15: H23: Product-Market Competition and Uncertainty Language

	(1)	(2)	(3)	(4)
	Industry FE		Firm FE	
	UncResCEO	UncPreCEO	UncResCEO	UncPreCEO
$z(\log(\text{TSIMM}))$	0.0008	**0.0304*****	0.0023	**0.0302*****
	(0.0014)	(0.0074)	(0.0075)	(0.0093)
UncQue	0.0011	**0.0398*****	0.0039	0.0084
	(0.0062)	(0.0100)	(0.0087)	(0.0073)
NegCall	0.0020	**0.2340*****	0.0046	**0.1698*****
	(0.0068)	(0.0201)	(0.0120)	(0.0140)
lnAssets	-0.0007	**-0.0335*****	-0.0101	0.0013
	(0.0007)	(0.0038)	(0.0071)	(0.0081)
TobinsQ	0.0001	-0.0007	0.0009	-0.0017
	(0.0013)	(0.0037)	(0.0025)	(0.0025)
ROA	0.0233	**0.0681****	0.0343	0.0241
	(0.0197)	(0.0314)	(0.0308)	(0.0254)
CashRatio	0.0065	**-0.0803****	0.0055	0.0022
	(0.0103)	(0.0360)	(0.0287)	(0.0306)
DivDummy	-0.0020	-0.0060	-0.0074	-0.0166
	(0.0028)	(0.0117)	(0.0083)	(0.0106)
FirmMat	0.0018	**0.0014****	0.0083	0.0003
	(0.0019)	(0.0007)	(0.0056)	(0.0004)
EarnVol	-0.0047	-0.0210	-0.0048	-0.0114
	(0.0087)	(0.0152)	(0.0121)	(0.0146)
UncPreCEO	**0.0111****		**0.0230****	
	(0.0044)		(0.0093)	
Industry FE	Yes	Yes		
Firm FE			Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	12,728	18,492	12,728	18,492
R^2	0.001	0.050	0.001	0.021
Adj. R^2	-0.002	0.048	-0.098	-0.068

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Unit of observation: firm-fiscal-year. R^2 includes absorbed fixed effects (not within-R^2).

Table 16: H24: US Economic Policy Uncertainty and Call Language Uncertainty

	(1) Industry + Cal. Year FE	(2)	(3) Firm + Cal. Year FE	(4)
	UncResCEO	UncPreCEO	UncResCEO	UncPreCEO
$\log(\text{US EPU})_t$	**0.0124****	**0.0211*****	**0.0123***	**0.0239*****
	(0.0075)	(0.0074)	(0.0075)	(0.0076)
UncQue	0.0011	**0.0164*****	0.0024	**0.0095*****
	(0.0035)	(0.0032)	(0.0040)	(0.0028)
NegCall	0.0006	**0.1247*****	0.0021	**0.1385*****
	(0.0056)	(0.0089)	(0.0073)	(0.0079)
lnAssets	0.0009	**-0.0143*****	0.0013	0.0038
	(0.0007)	(0.0020)	(0.0067)	(0.0063)
TobinsQ	-0.0001	0.0007	-0.0000	0.0007
	(0.0011)	(0.0019)	(0.0027)	(0.0018)
ROA	**0.0281***	**0.0367****	**0.0677****	**0.0467****
	(0.0149)	(0.0168)	(0.0268)	(0.0191)
CashRatio	0.0051	-0.0079	0.0345	0.0305
	(0.0087)	(0.0187)	(0.0256)	(0.0231)
DivDummy	**-0.0046*****	-0.0074	**-0.0142****	**-0.0157***
	(0.0009)	(0.0061)	(0.0072)	(0.0083)
FirmMat	**0.0023***	0.0009	0.0005	-0.0011
	(0.0012)	(0.0008)	(0.0047)	(0.0012)
EarnVol	**0.0268***	-0.0074	**0.0353****	-0.0083
	(0.0157)	(0.0086)	(0.0176)	(0.0119)
UncPreCEO	-0.0026		-0.0011	
	(0.0032)		(0.0045)	
Lagged_DV	**0.0202*****	**0.5176*****	**0.0187*****	**0.2626*****
	(0.0054)	(0.0131)	(0.0056)	(0.0089)
Industry FE	Yes	Yes		
Firm FE			Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	39,063	60,503	39,063	60,503
R^2	0.001	0.292	0.001	0.084
Adj. R^2	-0.000	0.291	-0.035	0.057

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) two-way clustered (firm, calendar quarter). Main sample (excludes financial and utility firms). Firms with fewer than 5 calls are excluded. R^2 includes absorbed fixed effects (not within-R^2).

Table 17: H24b: Global Economic Policy Uncertainty and Call Language Uncertainty

	(1)	(2)	(3)	(4)
	Industry + Cal. Year FE		Firm + Cal. Year FE	
	UncResCEO	UncPreCEO	UncResCEO	UncPreCEO
$\log(\text{GEPU})_t$	**0.0181****	**0.0271*****	**0.0187****	**0.0309*****
	(0.0100)	(0.0107)	(0.0101)	(0.0108)
UncQue	0.0011	**0.0164*****	0.0024	**0.0095*****
	(0.0035)	(0.0032)	(0.0039)	(0.0028)
NegCall	0.0006	**0.1247*****	0.0020	**0.1385*****
	(0.0056)	(0.0089)	(0.0072)	(0.0079)
lnAssets	0.0009	**-0.0143*****	0.0012	0.0037
	(0.0007)	(0.0020)	(0.0067)	(0.0063)
TobinsQ	-0.0001	0.0008	0.0000	0.0007
	(0.0011)	(0.0019)	(0.0027)	(0.0018)
ROA	**0.0280***	**0.0367****	**0.0673****	**0.0462****
	(0.0149)	(0.0168)	(0.0267)	(0.0191)
CashRatio	0.0051	-0.0080	0.0345	0.0304
	(0.0087)	(0.0188)	(0.0256)	(0.0231)
DivDummy	**-0.0047*****	-0.0074	**-0.0142****	**-0.0156***
	(0.0009)	(0.0061)	(0.0072)	(0.0083)
FirmMat	**0.0023***	0.0009	0.0006	-0.0011
	(0.0012)	(0.0008)	(0.0047)	(0.0012)
EarnVol	**0.0269***	-0.0074	**0.0354****	-0.0082
	(0.0157)	(0.0086)	(0.0176)	(0.0119)
UncPreCEO	-0.0026		-0.0011	
	(0.0032)		(0.0045)	
Lagged_DV	**0.0202*****	**0.5176*****	**0.0187*****	**0.2626*****
	(0.0054)	(0.0131)	(0.0056)	(0.0089)
Industry FE	Yes	Yes		
Firm FE			Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	39,063	60,503	39,063	60,503
R^2	0.001	0.292	0.001	0.084
Adj. R^2	-0.000	0.291	-0.035	0.057

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) two-way clustered (firm, calendar quarter). Main sample (excludes financial and utility firms). Firms with fewer than 5 calls are excluded. R^2 includes absorbed fixed effects (not within-R^2).

Table 18: H14c CEO 3-IV Decomp: ClarityCEO + UncResCEO (DWZ Eq.5 Full) + UncPreCEO (raw) and 25-Day Post-Call Bid-Ask Spread Level ($[0, +25]$, the call day plus 25 trading days)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Spread$_{25D,t}$						Spread$_{25D,t+1}$					
CEO Clarity (DWZ)	-0.0477	1.0617	0.2122	1.1056	0.1644	1.0547	-0.1886	0.5998	-0.0922	0.6828	-0.0431	0.5722
	(0.2145)	(0.4277)	(0.2227)	(0.4260)	(0.2103)	(0.4174)	(0.2370)	(0.4004)	(0.2399)	(0.4055)	(0.2206)	(0.3729)
CEO Residual Uncertainty (DWZ)	-0.0594	-0.0687	-0.0882	-0.1021	-0.0757	-0.0893	0.0825	0.0696	0.0742	0.0545	0.0984	0.0751
	(0.1068)	(0.1057)	(0.1023)	(0.1007)	(0.0994)	(0.0981)	(0.1416)	(0.1437)	(0.1407)	(0.1426)	(0.1339)	(0.1357)
CEO Pres Uncertainty	**0.1664****	**0.3496*****	0.0814	**0.2945*****	0.0108	**0.1644***	-0.0817	-0.0288	-0.1072	-0.0360	0.0170	0.1124
	(0.0946)	(0.1253)	(0.0956)	(0.1229)	(0.0919)	(0.1184)	(0.0926)	(0.1229)	(0.0926)	(0.1233)	(0.0872)	(0.1186)
lnAssets	**-0.8039*****	**-1.9098*****	**-0.6339*****	**-1.6825*****	**-0.6352*****	**-1.8277*****	**-0.8218*****	**-1.5191*****	**-0.7857*****	**-1.7292*****	**-0.6568*****	**-1.4181*****
	(0.0490)	(0.2000)	(0.0472)	(0.2156)	(0.0473)	(0.2158)	(0.0513)	(0.1951)	(0.0505)	(0.2130)	(0.0474)	(0.2017)
TobinsQ	**-0.2861*****	**-0.4785*****	**-0.2633*****	**-0.5518*****	**-0.2654*****	**-0.5883*****	**-0.2305*****	**-0.3215*****	**-0.2285*****	**-0.4803*****	**-0.2035*****	**-0.4289*****
	(0.0457)	(0.0900)	(0.0470)	(0.0963)	(0.0451)	(0.0956)	(0.0467)	(0.0850)	(0.0478)	(0.1002)	(0.0433)	(0.0935)
ROA	**-9.4469*****	**-11.6430*****	**-5.7059*****	**-7.6844*****	**-5.7511*****	**-7.8517*****	**-9.9256*****	**-12.8443*****	**-8.8744*****	**-11.9197*****	**-7.6375*****	**-10.5867*****
	(1.0201)	(1.3867)	(0.9822)	(1.2654)	(0.9633)	(1.2490)	(1.0798)	(1.2992)	(1.0722)	(1.2732)	(0.9869)	(1.1921)
Leverage	**0.3875***	**1.8330*****	-0.1126	0.6850	-0.0542	0.8026	0.2924	0.5282	0.1338	0.2221	0.0592	0.1928
	(0.2088)	(0.5143)	(0.2434)	(0.5013)	(0.2264)	(0.4954)	(0.2224)	(0.5516)	(0.2248)	(0.5596)	(0.2111)	(0.5258)
Capex	-0.8001	0.9540	**-2.3622****	-1.5271	**-2.0790***	-1.2897	**-1.6463***	-3.2121	**-2.1561****	**-4.3379***	-1.4770	-2.4356
	(1.1109)	(2.9126)	(1.1044)	(2.8061)	(1.0845)	(2.7998)	(0.9620)	(2.2020)	(0.9823)	(2.2446)	(0.9011)	(2.1565)
DivDummy	0.1043	0.1429	**0.3329*****	0.2549	**0.3156*****	0.2723	0.0240	0.1007	0.1012	0.1454	0.1145	0.1739
	(0.0893)	(0.2020)	(0.0896)	(0.1888)	(0.0855)	(0.1867)	(0.0887)	(0.2095)	(0.0889)	(0.2074)	(0.0825)	(0.1975)
sCFO	**0.6700****	**0.8405****	0.4045	**0.6825***	0.4084	**0.6882****	0.0871	**0.3644***	0.0040	0.3165	-0.0840	0.2479
	(0.3405)	(0.3472)	(0.4604)	(0.3789)	(0.4126)	(0.3375)	(0.1759)	(0.1877)	(0.2174)	(0.2055)	(0.1937)	(0.1738)
Lagged_DV	**0.7547*****	**0.6569*****	**0.7140*****	**0.6180*****	**0.7310*****	**0.6318*****	**0.7540*****	**0.6361*****	**0.7360*****	**0.6135*****	**0.7676*****	**0.6518*****
	(0.0150)	(0.0186)	(0.0167)	(0.0199)	(0.0168)	(0.0208)	(0.0156)	(0.0185)	(0.0173)	(0.0205)	(0.0162)	(0.0197)
StockPrice			-0.0011	-0.0033	0.0002	0.0019			-0.0001	**0.0101*****	-0.0016	**0.0047****
			(0.0014)	(0.0024)	(0.0013)	(0.0024)			(0.0013)	(0.0024)	(0.0012)	(0.0022)
Turnover			**-24.1012*****	**-23.8647*****	**-20.7840*****	**-20.3411*****			**-7.9396*****	**-5.5050*****	**-7.1303*****	**-4.7073****
			(1.7508)	(2.2119)	(1.7290)	(2.1791)			(1.5411)	(2.0904)	(1.4699)	(1.9633)
DailyVola			**0.1229*****	**0.1393*****	**0.1075*****	**0.1257*****			**0.0404*****	**0.0520*****	**0.0396*****	**0.0441*****
			(0.0072)	(0.0081)	(0.0082)	(0.0099)			(0.0062)	(0.0068)	(0.0069)	(0.0081)
AbsSurpDec			**-0.0919*****	-0.0241	**-0.0756*****	-0.0093			-0.0050	0.0119	-0.0174	-0.0012
			(0.0265)	(0.0300)	(0.0253)	(0.0286)			(0.0278)	(0.0304)	(0.0261)	(0.0285)
Extended Controls			Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes
Industry FE	Yes		Yes		Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes		Yes		Yes
Year FE	Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes		
Year-Quarter FE					Yes	Yes					Yes	Yes
N	42,625	42,625	42,625	42,625	42,625	42,625	43,049	43,049	43,049	43,049	43,049	43,049
R^2	0.703	0.517	0.721	0.551	0.730	0.548	0.700	0.499	0.702	0.504	0.723	0.523
Adj. R^2	0.703	0.501	0.721	0.535	0.730	0.532	0.700	0.482	0.702	0.487	0.723	0.506

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). DWZ Full decomposition: Clarity = -CEO FE; UncRes = call-level residual. R^2 includes absorbed fixed effects (not within-R^2). Spread$_{25D}$ values rescaled by 10^4 (basis points) for readability; multiply coefficients by 10^{-4} to recover raw fraction units.

Table 19: H18 CEO 3-IV Decomp: ClarityCEO + UncResCEO (DWZ Eq.5 Full) + UncPreCEO (raw) and SEC Conference-Call Comment Letter Receipt (LPM, $[call_t, call_{t+1}]$ window)

	(1)	(2)	(3)	(4)	(5)	(6)
	CCCL					
CEO Clarity (DWZ)	0.0059	0.0039	0.0062	0.0040	0.0058	0.0040
	(0.0015)	(0.0030)	(0.0015)	(0.0030)	(0.0015)	(0.0030)
CEO Residual Uncertainty (DWZ)	-0.0003	-0.0003	-0.0003	-0.0003	-0.0004	-0.0004
	(0.0009)	(0.0009)	(0.0009)	(0.0009)	(0.0009)	(0.0009)
CEO Pres Uncertainty	0.0008	**0.0014***	**0.0016****	**0.0014***	0.0007	**0.0013***
	(0.0008)	(0.0009)	(0.0007)	(0.0009)	(0.0008)	(0.0009)
lnAssets	**-0.0006****	**0.0018***	0.0001	0.0017	**-0.0006****	**0.0019***
	(0.0002)	(0.0011)	(0.0001)	(0.0010)	(0.0003)	(0.0011)
TobinsQ	-0.0003	0.0002	-0.0001	0.0002	-0.0002	0.0004
	(0.0003)	(0.0004)	(0.0003)	(0.0004)	(0.0003)	(0.0004)
ROA	0.0006	0.0012	-0.0014	-0.0006	-0.0006	-0.0004
	(0.0034)	(0.0052)	(0.0052)	(0.0062)	(0.0051)	(0.0062)
Leverage	-0.0008	-0.0019	-0.0009	-0.0011	-0.0009	-0.0012
	(0.0015)	(0.0034)	(0.0015)	(0.0035)	(0.0015)	(0.0035)
Capex	-0.0029	0.0106	0.0004	0.0111	-0.0033	0.0107
	(0.0063)	(0.0104)	(0.0064)	(0.0104)	(0.0066)	(0.0103)
CashRatio	-0.0014	**-0.0075***	0.0009	**-0.0080***	-0.0014	**-0.0082***
	(0.0027)	(0.0044)	(0.0026)	(0.0044)	(0.0028)	(0.0044)
DivDummy	-0.0005	0.0003	-0.0005	0.0002	-0.0005	0.0002
	(0.0008)	(0.0015)	(0.0008)	(0.0015)	(0.0008)	(0.0015)
sCFO	-0.0007	-0.0020	-0.0004	-0.0019	-0.0007	-0.0019
	(0.0005)	(0.0024)	(0.0004)	(0.0024)	(0.0005)	(0.0024)
Lagged_DV	**-0.0079*****	**-0.0409*****	**-0.0077*****	**-0.0410*****	**-0.0074*****	**-0.0405*****
	(0.0008)	(0.0027)	(0.0007)	(0.0027)	(0.0008)	(0.0028)
SalesGrowth			-0.0002	-0.0013	-0.0001	-0.0011
			(0.0008)	(0.0011)	(0.0008)	(0.0011)
RDSales			-0.0003	-0.0002	-0.0003	-0.0001
			(0.0002)	(0.0002)	(0.0002)	(0.0002)
CashFlowAt			0.0025	0.0043	0.0009	0.0037
			(0.0050)	(0.0064)	(0.0050)	(0.0064)
DailyVola			0.0000	**-0.0000***	0.0000	-0.0000
			(0.0000)	(0.0000)	(0.0000)	(0.0000)
Extended Controls			Yes	Yes	Yes	Yes
Industry FE	Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes
Year FE	Yes	Yes	Yes	Yes		
Year-Quarter FE					Yes	Yes
N	44,113	44,113	44,096	44,096	44,096	44,096
R^2	0.001	0.002	0.000	0.002	0.001	0.002
Adj. R^2	-0.000	-0.032	-0.001	-0.032	-0.001	-0.033

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). DWZ Full decomposition: Clarity = -CEO FE; UncRes = call-level residual. R^2 includes absorbed fixed effects (not within-R^2).

Table 20: Phase E CEO sudden-death DiD: ΔCash around exogenous CEO shock; PSM 1:1 nearest-neighbor controls on Bates (2009) covariates at $t-1$

FE	(1) Industry Ghafoor (2023)	(2) Firm	(3) Ind+YQ	(4) Firm+YQ
Treated	+0.0352***		+0.0326***	
	(0.0129)		(0.0123)	
Post	+0.0212**	+0.0033	+0.0384***	+0.0264*
	(0.0094)	(0.0086)	(0.0096)	(0.0140)
Treated × Post (ATT)	-0.0178	-0.0090	-0.0162	-0.0053
	(0.0163)	(0.0141)	(0.0138)	(0.0127)
N (firm-quarters)	338	338	338	338
R^2	0.406	0.320	0.420	0.320
N (firms)	16	16	16	16

8 viable treated firms (CEO sudden death 2002-2018; ±12 quarter coverage; ≥5 pre-event UncAnsCEO obs) matched 1:1 to non-event controls via propensity-score nearest-neighbor on lnAssets, Leverage, ROA, DivDummy, sCFO at $t-1$. Window: ±12 fiscal quarters around death event. Col (1) replicates Ghafoor (2023) §4.5 specification (industry FE; firm-clustered SE). Standard errors clustered at firm level; with 16 clusters (8 treated + 8 PSM-matched controls) asymptotic cluster-robust inference may be downward-biased; results interpreted as suggestive (advisor-recommended). Parallel-trends placebo (Treated × Pre-4q): β=+0.0266, p=0.183; Treated × Pre-8q: β=+0.0210, p=0.146. Both insignificant supports DiD identification. PSM balance (std. diff): lnAssets=+0.04, Leverage=-0.08, ROA=-0.04, DivDummy=+0.00, sCFO=+0.19 (all | < 0.25|, Rosenbaum-Rubin standard). Coefficients on Lagged DV and other controls suppressed for space. ATT p-values are one-tailed (Ghafoor 2023 negative prior). $^{*}/^{**}/^{***}$ denote $p < 0.10/0.05/0.01$.

Table 21: DWZ §6 First-Difference Replication: CEO Turnover and Cash Holdings

FE	(1) DWZ §6 FF12	(2) + Bates FF12	(3) Ind-adj demeaned
Δ ClarityCEO	-0.0183**	-0.0172*	-0.0182**
	(0.0109)	(0.0107)	(0.0108)
Δ NegCall	-0.0390**	-0.0369**	-0.0387**
	(0.0162)	(0.0163)	(0.0161)
Δ UncPreCEO	-0.0002	-0.0053	-0.0002
	(0.0156)	(0.0151)	(0.0156)
Δ UncQue	+0.0221	+0.0194	+0.0223
	(0.0157)	(0.0156)	(0.0156)
Δ lnAssets	-0.0321***	-0.0280***	-0.0322***
	(0.0072)	(0.0074)	(0.0072)
Δ ROA	+0.0645	+0.0532	+0.0653
	(0.0592)	(0.0578)	(0.0591)
Δ Leverage		-0.0686**	
		(0.0321)	
Δ DivDummy		+0.0118	
		(0.0129)	
Δ sCFO		+0.2054***	
		(0.0766)	
N (turnover pairs)	659	659	659
R^2	0.108	0.128	0.059

Sample: 661 CEO turnover pairs in F1D Main-industry firms 2002Q1–2018Q4. One observation per turnover pair; variables averaged over each CEO's tenure at the firm. WLS weighted by harmonic-mean of quarter counts. Replicates Dzielinski, Wagner & Zeckhauser (2021) §6 Equation 7 first-difference design, adapted from Tobin's Q to cash holdings. One observation per CEO turnover pair (gvkey, old, new); variables averaged over each CEO's tenure at the firm; Δ = New − Old. Estimated by weighted least squares with weights $w = (n_{\text{old}} \cdot n_{\text{new}})/(n_{\text{old}} + n_{\text{new}})$ per DWZ verbatim. Robust (HC1) standard errors. Sample filters per DWZ §6 verbatim: gap between old CEO last call and new CEO first call ≤ 120 days; both CEOs ≥ 5 calls at firm; both CEOs have valid ClarityCEO. Deviations from DWZ verbatim: FF12 industry FE (vs FF48); intangibles ratio omitted (not in panel); sample = F1D Main industries (FF12 ∉ {Finance, Utility}). **Endogeneity caveat** (§4.3 disclosure): DWZ themselves note this design "does not completely eliminate endogeneity concerns" (§6); because cash is a CEO-discretion choice variable, the endogeneity profile here is more concerning than in DWZ's Tobin's Q application. This table is a robustness companion to the Phase E sudden-death DiD (Table 20); forced/voluntary turnover not separated. p-value on Δ ClarityCEO is one-tailed (H1 prior: clarity ↑⇒ cash ↓); other p-values two-tailed. $^{*}/^{**}/^{***}$ denote $p < 0.10/0.05/0.01$.

Table 22: Lewbel (2012) Heteroskedasticity-Based IV: Cash Holdings

	(1) OLS	(2) 2SLS-Lewbel	(3) + extended
UncResCEO	+0.0021**	+0.0100	+0.0101
	(0.0010)	(0.0083)	(0.0085)
N	43,471	43,471	43,454
N firms	1,423	1,423	1,423
R^2	0.863	0.863	0.864
First-stage F	n/a	20.4	41.3
Sargan p	n/a	0.919	0.009
Wu-Hausman p	n/a	0.241	0.159

Sample: HC Main panel — F1D firm-quarters in Main industries (FF12 ∉ {Finance, Utility}) 2002Q1–2018Q4. Identical to H1.ceo2 main-panel HC sample. Lewbel (2012) heteroskedasticity-based IV identifies β on UncResCEO via internal instruments $Z_j = (W_j - \bar{W}_j) \cdot \hat{\varepsilon}_X^{\text{1st-stage}}$ for each W_j with significant Pesaran-Taylor heteroskedasticity in the first-stage residual. Col (1) OLS baseline replicates the H1.ceo2 main panel HC specification on the same sample. Cols (2)-(3) are 2SLS with Lewbel-manufactured instruments. Pesaran-Taylor p-values ($W_j \rightarrow$ heteroskedasticity in $\hat{\varepsilon}_X$): Leverage=0.000, lnAssets=0.000, TobinsQ=0.000, ROA=0.004, Capex=0.000, DivDummy=0.733, sCFO=0.119, CashRatio_lag=0.017. Standard errors firm-clustered. **Identification scope (§4.3 disclosure):** Lewbel addresses omitted TIME-VARYING confounders within firm spells (which neither firm-FE nor manager-FE absorbs). For reverse-causality concerns (firms hoard cash → CEO talks more uncertainly about that position), the design is weaker than Phase E sudden-death (Table 20); the three designs together cover three different identification threats rather than converge on a single coefficient. p-value on UncResCEO is one-tailed (H1 prior: UncResCEO ↑⇒ Cash ↑). $^{*}/^{**}/^{***}$ denote $p < 0.10/0.05/0.01$.